

Q1) Interplanar spacing and angle in 3D Bravais lattices: (30 pts)

- Starting from the definition of the reciprocal space lattice vector, derive the interplanar distance d_{hkl} equation with respect to (hkl) planes in the cubic system with lattice constant a .
- Starting from the definition of the reciprocal space lattice vector, express the interplanar distance d_{hkl} between (hkl) planes in the triclinic system.
- Compute the angle between the (321) and (112) planes in
 - $\{1, 1, 1, 90, 90, 90\}$
 - $\{1, 1, 2, 90, 90, 90\}$
 - $\{1, 1, 2, 90, 75, 85\}$

Q2) Identify family of planes: (10 pts)

What is the family of planes $\{hkl\}$ corresponding with an interplanar spacing of $d = 1.246 \text{ \AA}$ in nickel (Ni) with $a = 3.524 \text{ \AA}$? Hint: the solution requires identifying several (hkl) planes that fulfill the interplanar spacing and concluding the family of planes they belong to.

Q3) Reciprocal space lattice: (25 pts)

Starting with a tetragonal P lattice, with lattice parameters a and c , **calculate** the reciprocal lattice vectors, **draw** the reciprocal Bravais lattice, and **indicate** the lattice vectors $\mathbf{a}^*$, $\mathbf{b}^*$ and $\mathbf{c}^*$. **Draw** a 3D lattice corresponding to $2 \times 2 \times 2$ unit cells (8 unit cells in total). Place the origin at the center and using the reciprocal lattice vector and the addition rule, **index** all reciprocal lattice points.

Q4) Reciprocal metric tensor and x-ray diffraction: (35 pts)

In 2004, NASA's Mars rover Spirit examined the Columbia Hills region of Mars and found the mineral goethite. Goethite is known to form only in the presence of water and the discovery was taken as evidence for water activity on Mars. Its chemical composition is iron(III) oxide-hydroxide, its space group is Pbnm, and the values of its lattice constants are $a = 0.4596$, $b = 0.9957$, and $c = 0.3021 \text{ nm}$. Determine the following:



- (a) The reciprocal lattice parameters of goethite.
- (b) Draw a 3D lattice corresponding to $2 \times 2 \times 2$ unit cells, placing the origin at the center and index all reciprocal lattice points.
- (c) The general reciprocal metric tensor of the crystal system goethite belongs to.
- (d) The general distance between any (hkl) planes for the crystal system.
- (e) The specific reciprocal metric tensor for goethite.
- (f) The values for the interplanar spacing between (020) , (200) , and (111) planes in goethite.
- (g) The angle between the plane normal (101) and $(\bar{1}01)$ in goethite.